HUMORIX AI

Affective & Multimodal Emotion Intelligence

AMI · AEI · CBT · SATVIC WELLNESS

MASTER INTELLIGENCE DOCUMENT

Complete Architecture · Clinical Protocols · Emotion Science · Holistic Wellness System

MULTIMODAL	EMOTION AI	MENTAL HEALTH	CBT ENGINE	SATVIC WELLNESS	OPEN SOURCE

Humorix AI is an emotionally intelligent system that detects, understands, and responds to human emotional states through facial recognition, voice analysis, and natural language processing — delivering evidence-based therapeutic support, holistic lifestyle guidance, and responsible crisis escalation in a single unified platform.

Version	1.0 — Master Edition
Scope	Complete AMI/AEI Architecture + Clinical Protocols + Satvic Wellness
Technology Stack	Python · PyTorch · OpenCV · MediaPipe · FER · Transformers · Streamlit

Disclaimer	Supportive tool only. Not a substitute for licensed professional care.

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PART I

WHAT IS HUMORIX AI? — VISION & PHILOSOPHY

1.1 Definition: AMI & AEI

AMI (Affective Machine Intelligence)	An AI system capable of recognising, interpreting, and responding to the full spectrum of human emotional states. AMI goes beyond passive analysis — it actively adapts its behaviour based on what it perceives emotionally, creating a feedback loop between the human and the machine.
AEI (Affective Emotion Intelligence)	The specific sub-domain of AMI focused on the emotional reasoning layer — the ability to not just detect an emotion label ('sad') but to understand context, severity, disorder risk, and the appropriate therapeutic response. AEI is what transforms a detection model into a supportive intelligence.
Humorix = AMI + AEI	Humorix combines both: it detects via AMI (multimodal sensors) and then reasons via AEI (clinical protocols, CBT frameworks, and holistic wellness knowledge) to generate personalised, empathetic support responses.

1.2 Core Mission & Goals

Scalable Mental Health Support	Bridge the gap between the rising demand for mental health care and the limited global supply of professionals — available 24/7, in any location, at any scale.
Multimodal Understanding	Understand the human not just through words but through face, voice, and language simultaneously — just as a skilled human therapist would.
Evidence-Based Interventions	Deliver only clinically grounded techniques (CBT, DBT principles, mindfulness, ERP) — no generic advice, no pseudoscience.
Holistic Wellness	Integrate the Satvic lifestyle philosophy to address the mind-body connection — because mental health cannot be separated from physical nourishment.

Responsible AI	Maintain strict ethical boundaries: never diagnose, always disclaim, always escalate in crisis situations, always refer to professionals.

1.3 Why It Matters — The Mental Health Gap

Mental health conditions affect over 1 billion people worldwide. The WHO estimates a treatment gap of 70–90% in low-to-middle income countries. In India alone, there are fewer than 0.3 psychiatrists per 100,000 people. Stigma, cost, and geographic barriers prevent millions from accessing care. Humorix AI is designed as a first-response, always-available companion — not a replacement for therapy, but a bridge toward it.

■■ Humorix does not diagnose or treat mental illness. It provides evidence-based supportive techniques and always escalates to professionals when severity thresholds are crossed.

PART II

EMOTION SCIENCE FOUNDATION

2.1 What Is Emotion Perception?

Emotion perception is the capacity to recognise and identify emotions in others. It is a sophisticated process where our sensory systems (sight, sound, smell) take in physical cues — like a scowl or a trembling voice — and the brain translates them into a mental representation of that person's internal feelings. It acts as a bridge between a physical observation and a psychological understanding.

2.2 Multi-Sensory Emotion Detection

Modality	How It Works & Examples
Visual (Primary)	Facial expressions (muscle movements — Action Units), body posture, micro-expressions, environmental context (tears at a wedding vs. funeral carry very different meaning).
Auditory	Prosody — pitch, duration, loudness, speech rate, tremor. Music can trigger emotional recognition without any language cues at all.
Olfactory	Scent associations linked to emotional memories. Certain smells reliably trigger mood changes through learned associations stored in the limbic system.

2.3 Three Major Theories of Emotion

Theory	Core Principle	Example
James-Lange	Physical response FIRST, then emotion.	You run and sweat, so you conclude you are afraid.
Cannon-Bard	Physical AND emotional responses happen simultaneously, independently.	You feel fear AND start sweating at the exact same moment — neither causes the other.
Two-Factor (Schachter-Singer)	Physical arousal first, then cognitive labelling using context.	Heart races → you see a bear → you label the sensation as 'fear'. Same arousal could be labelled excitement.

■ Humorix's fusion engine implicitly operates on Two-Factor principles: raw arousal signals (voice, face) are combined with textual context to arrive at a labelled emotional state.

2.4 The Brain's Emotional Hardware

Fusiform Face Area (FFA)	Specialised region in the temporal lobe that identifies and processes human faces. Damage here causes prosopagnosia (face blindness). FFA activity is what Humorix's facial detection layer computationally mimics.
Amygdala	The brain's alarm system. Crucial for detecting threats and directing attention to emotionally charged stimuli, especially fear and anger. Overactive in PTSD and anxiety disorders.
HPA Axis (Hypothalamic-Pituitary-Adrenal)	Manages the stress response — releasing cortisol and adrenaline to prepare the body to react. Chronic activation leads to anxiety, sleep disruption, and immune suppression.
Prefrontal Cortex	The brain's 'brakes'. Regulates amygdala reactivity, enables emotional context assessment, and supports cognitive reappraisal — the foundation of CBT techniques.

2.5 Cultural & Disorder Differences

Western vs. Eastern Culture	Research shows Western cultures tend to focus on the MOUTH to judge emotion; Eastern cultures focus more on the EYES. This affects how facial AUs (Action Units) should be weighted — a reason Humorix includes bias mitigation protocols across demographics.
Autism Spectrum	Reduced ability to decode subtle facial expressions. Standard emotion recognition models may fail here. Humorix includes adaptive protocols and higher reliance on verbal/text signals for these users.
Schizophrenia	Impaired facial affect recognition and flat affect production. Can cause misclassification in both directions.
Anxiety & PTSD	Threat bias — individuals perceive anger or danger in neutral faces more rapidly than others. Humorix's PTSD and GAD modules account for this perceptual amplification.

2.6 Research Measurement Tools Used in Emotion Science

Dot Probe Task	Measures attentional bias toward emotional stimuli. A dot replaces an emotional image — faster detection indicates the attention was already directed there.
Modified Stroop Task	Participant names the INK COLOR of an emotionally loaded word (e.g., 'WAR'). Slower reaction = emotional meaning is competing for cognitive resources.
Action Units (FACS)	Facial Action Coding System — decomposes facial expressions into individual muscle movements. The scientific foundation behind FER systems.

PART III

SYSTEM ARCHITECTURE

3.1 Five-Layer Architecture

Layer	Components	Function
1. INPUT LAYER	Webcam (real-time video) · Microphone (audio stream) · Text input (keyboard/chat)	Raw human signal capture
2. PROCESSING LAYER	OpenCV + MediaPipe (face detection) · Librosa (audio features) · Transformers NLP	Signal preprocessing & feature extraction
3. EMOTION LAYER	FER / DeepFace (facial AU classification) · ML models (voice emotion) · Sentiment classifier (text)	Emotion state classification
4. DECISION LAYER	Fusion Engine · InterventionEngine · Satvic Knowledge Base	Reasoning, weighting, intervention selection
5. OUTPUT LAYER	Text response · Voice feedback · Streamlit UI dashboard · Crisis escalation pathway	Human-facing support delivery

3.2 Facial Emotion Recognition (FER)

Facial emotion recognition is the cornerstone of Humorix's visual modality. Deep learning models are trained on curated datasets to identify micro-expressions — involuntary muscle movements lasting as little as 1/25th of a second — that reveal genuine emotional states even when consciously suppressed.

Libraries Used	OpenCV (real-time capture) · MediaPipe (landmark detection) · FER library · DeepFace
Key Features Detected	Eyebrow movement (corrugator, frontalis) · Eye openness (levator) · Mouth curvature (zygomaticus, depressor) · Nose wrinkling · Head pose
Performance Target	30 FPS with GPU acceleration — real-time without perceptible lag
Baseline Emotions	Happy · Sad · Angry · Fearful · Disgusted · Surprised · Neutral (Ekman's 7 Universal Emotions)

Confidence Scoring	Each emotion label carries a probability score (0.0–1.0). Low confidence triggers multi-modal confirmation before acting.

3.3 Voice Emotion Detection

Voice provides a parallel emotional channel — one that is harder to consciously mask than facial expression. Librosa extracts acoustic features from the audio stream, which are then fed into trained classifiers.

Library	Librosa (Python audio analysis)
Features Extracted	MFCCs (Mel-Frequency Cepstral Coefficients) — voice timbre fingerprint Spectral Contrast — frequency range dynamics Pitch & Tone — fundamental frequency contour Speech Rate — words per minute, pause patterns Tremor / Jitter — instability indicating stress or emotion
Detected States	Stress · Anxiety · Calmness · Sadness · Anger · Elevated mood (mania signal)
Panic Detection	Rapid speech + elevated pitch + irregular rhythm = panic disorder signal

3.4 Text-Based Sentiment Analysis (NLP)

Model Type	Transformer-based large language models (BERT / RoBERTa family)
Classification Categories	Positive · Negative · Neutral · Distress · Hopeless · Anger · Intrusive thoughts · Safety concern
Distress Pattern Detection	Negative self-talk · Hopelessness · Social isolation language · Repetitive thought loops (OCD signal) · Worthlessness statements (depression signal)
Intent Analysis	Differentiates venting from seeking advice from crisis disclosure

3.5 Multimodal Fusion Engine

The Fusion Engine is the intelligence core of Humorix — the component that combines the three modality signals into a single, reliable emotional state assessment. Single-modality systems are fundamentally limited: a person can have a neutral face while typing distressed text. The fusion layer resolves these contradictions.

Fusion Method	Weighted averaging with rule-based priority overrides
Base Weights	Facial: 0.45 · Voice: 0.30 · Text: 0.25 (adjustable per context)

Priority Rules	Crisis language in text overrides all other signals (safety first) Panic voice pattern + fearful face = panic disorder path regardless of neutral text Low facial confidence → increase text/voice weight automatically
Output	fused_emotion · severity (none/mild/moderate/severe/crisis) · disorder_risk dict · overall_confidence score

■ The fusion engine produces a structured dict that the InterventionEngine consumes. This clean interface allows each module to be independently upgraded without breaking the pipeline.

3.6 Training Datasets

Dataset	Size / Content	Primary Use
FER2013	35,887 labelled facial images	Core facial emotion classification
AffectNet	~1 million images, 8 emotion categories	Large-scale robustness & diversity
CK+ (Extended Cohn-Kanade)	Facial expression sequences (posed & spontaneous)	Temporal expression analysis
DAIC-WOZ	Clinical depression interviews with PHQ-8 scores	Depression signal training
IEMOCAP	Interactive emotional dyadic motion capture database	Voice emotion training

Training Frameworks	PyTorch · TensorFlow · Hugging Face Transformers
Mobile Export	ONNX conversion for mobile/edge deployment

3.7 Deployment Modes

Local (Default)	Python + Streamlit — full control, private, runs on local GPU/CPU
Cloud Scale	Docker containerisation → AWS / GCP / Azure deployment for multi-user scale
Mobile	ONNX model conversion → React Native or Flutter frontend
Real-Time Performance	30 FPS with GPU · 5–10 FPS CPU-only fallback
Clinical Integration	REST API layer allows integration with EHR/EMR systems or telehealth platforms

PART IV

CLINICAL INTERVENTION ENGINE

4.1 The DVITE Protocol

Every interaction in Humorix follows the DVITE protocol — a five-stage pipeline derived from Cognitive Behavioural Therapy principles, adapted for AI-delivered support.

① DETECT	② VALIDATE	③ INTERVENE	④ TRACK	⑤ ESCALATE

① DETECT	Fusion Engine identifies emotional state, severity, and disorder risk probability.
② VALIDATE	AI delivers a clinically worded validation statement — normalises the experience without minimising it. Never toxic positivity.
③ INTERVENE	Evidence-based technique selected from disorder-specific library based on severity (mild / moderate / severe). Examples: 5-4-3-2-1 grounding, 4-7-8 breathing, thought records, ERP ladders.
④ TRACK	User self-reports intensity on 1–10 scale. System records scores across session to calculate improving / stable / worsening trend.
⑤ ESCALATE	If severity or intensity exceeds threshold, system automatically provides Indian + international crisis helpline resources and urges professional consultation.

4.2 Disorder Module Library

Disorder	Full Name	Threshold & Key Techniques
GAD	Generalised Anxiety Disorder	Escalate: 7/10 5-4-3-2-1 grounding · 4-7-8 breathing · worry thought challenging · PMR
Panic Disorder	Panic & Agoraphobia	Escalate: 8/10 Box breathing · DARE Response · extended exhale technique
Depression / MDD	Major Depressive Disorder	Escalate: 7/10 Behavioural activation · pleasure prediction · thought records · social engagement

PTSD	Post-Traumatic Stress	Escalate: 7/10 Grounding anchoring · safe place visualisation · titrated exposure · window of tolerance
Bipolar — Mania	Manic/Hypomanic Episodes	Escalate: 7/10 Sleep regulation · 48-hour decision rule · energy channelling
OCD	Obsessive-Compulsive Disorder	Escalate: 8/10 Thought defusion · ERP ladder · compulsion delay technique
Eating Disorder	Anorexia / Bulimia / BED	Escalate: 6/10 Body neutrality · mindful eating · thought challenging
ADHD	Attention Deficit Hyperactivity	Escalate: 8/10 Body doubling · 2-minute rule · Pomodoro · externalise everything

4.2a Detailed: GAD — Generalised Anxiety Disorder

Detect Signal	Facial tension, eyebrow raise, rapid blinking + worrying language patterns in text
Mild Interventions	• 5-4-3-2-1 Grounding: name 5 things visible, 4 touchable, 3 audible, 2 smellable, 1 tasteable • Extended exhale breathing: in 4 counts, hold 4, out 6 (activates parasympathetic NS) • Write the worry → assess controllability → practice sitting with uncertainty
Moderate Interventions	• 4-7-8 breathing: inhale 4, hold 7, exhale 8 — repeat 3 cycles • Cognitive challenging: probability estimation of feared outcome + evidence weighing • Progressive Muscle Relaxation from feet to face
Severe	Professional CBT referral + KIRAN helpline + diaphragmatic breathing immediate stabilisation

4.2b Detailed: Depression / MDD

Detect Signal	Low affect (flat face), slower speech, reduced prosody range, hopelessness language
Mild Interventions	• Behavioural Activation: identify one 5-minute activity with any positive valence • Gratitude micro-practice: one item, however small (rewires negativity bias) • Opposite action: gentle movement vs. urge to stay still
Moderate Interventions	• Pleasure prediction experiment: predict enjoyment (1-10) → do activity → compare actual rating • Full thought record: thought → evidence for → evidence against → balanced alternative • Social connection micro-action: send one message to one person

Severe	Direct safety inquiry + professional referral + KIRAN / iCall / Vandrevala resources

4.2c Detailed: PTSD

Detect Signal	Avoidance language, hypervigilance descriptors, intrusion reports, startle responses
Mild	• Grounding anchor: press hand on solid surface, feel temperature — present-moment anchor • Safe place visualisation: 2 minutes in a real or imagined completely safe environment
Moderate	• Window of tolerance: name 'I am having a trauma response' — activates PFC, reduces amygdala • Titrated exposure: 30-second controlled acknowledgement of the memory, then return to grounding
Severe	EMDR + trauma-focused CBT referral + trauma-informed therapist finder support

4.3 Severity Levels & Escalation Logic

Severity	Criteria	Response Action
none	Baseline / positive state	Psychoeducation, check-in, wellness tips
mild	Subclinical distress	Mild interventions, grounding, validation
moderate	Significant impairment	Moderate CBT techniques, tracking, soft professional referral
severe	High distress + risk markers	Severe interventions + hard professional referral + resources listed
crisis	Immediate safety concern	CRISIS PROTOCOL — helplines foregrounded immediately, all other content secondary

4.4 Crisis Protocol & Indian Resources

■■ When severity == 'crisis', the system immediately validates, provides breathing stabilisation, and foregrounds crisis resources. No diagnosis, no delay.

iCall (TISS)	9152987821 — Free, trained counsellors
Vandrevala Foundation	1860-2662-345 — 24/7 multilingual crisis support
KIRAN Mental Health Helpline	1800-599-0019 — Free, 24/7, Government of India

Snehi	044-24640050 — Suicide prevention
Crisis Text Line (International)	Text HOME to 741741
988 Lifeline (US)	Call or text 988
Samaritans (UK)	116 123
Global Directory	findahelpline.com

4.5 Session Tracking & Trend Analysis

Humorix tracks user self-reported intensity scores (1–10) across the session for each disorder. After 3 or more data points, the trend engine computes a direction:

Trend	Criterion	System Response
Improving	Recent avg < Earlier avg by >1 point	Affirm progress, maintain intervention
Stable	Recent avg within ±1 of earlier avg	Continue current approach, check-in depth
Worsening	Recent avg > Earlier avg by >1 point	Escalate to next severity tier, strengthen referral

PART V

SATVIC WELLNESS INTELLIGENCE

5.1 Satvic Philosophy & Food Classification

Rooted in Chapter 17 of the Bhagavad Gita and systematised by Subah Saraf's Satvic Food Book, this philosophy classifies all food and lifestyle choices into three modes — Gunas — that directly affect mental and emotional states. Humorix integrates this as its holistic wellness layer.

Mode (Guna)	Food Characteristics	Mental Effect
SATVIC (Mode of Goodness)	Fresh, living, plant-based, water-rich. Cooked minimally.	Purity · Clarity · Bliss · Compassion · Self-control · Fearlessness
RAJASIK (Mode of Passion)	Spicy, stimulating, fast food, processed, caffeinated.	Arrogance · Restlessness · Anxiety · Anger · Uncontrollable desire
TAMASIK (Mode of Ignorance)	Stale, dead, packaged, overcooked, fermented, meat.	Laziness · Depression · Lethargy · Ignorance · Illusion

The 4 Satvic Food Principles — LWPW

LIVING	WHOLESOME	PLANT-BASED	WATER-RICH

LIVING	Food direct from farm to kitchen. Nothing packaged, tinned, bottled, or canned.
WHOLESOME	Unprocessed, unrefined. Whole grains, dates, jaggery, unpolished rice.
PLANT-BASED	From plants and trees only. No meat, fish, eggs, or dairy.
WATER-RICH	High water content dominates: fruits, vegetables, leafy greens.

5.2 The 9 Satvic Food Laws

Law	Avoid	Eat
1. No Dead Foods	Avoid: Packaged snacks, chips, namkeen, sauces, bottled dressings	Eat: Fresh fruits, vegetables, grains, nuts, seeds — farm to kitchen
2. No Refined Foods	Avoid: White sugar, white flour, white rice, refined oils	Eat: Dates, jaggery, millets/whole wheat, brown/red rice, cold-pressed oils

3. No Animal Products	Avoid: Meat, fish, eggs, dairy, paneer, ghee, butter	Eat: Homemade coconut milk, almond milk, cashew cheese, nut butter
4. Less Grain, More Veggies	Avoid: Grain-heavy plates (especially in healing phase)	Eat: At least 2x more vegetables than grains or pulses
5. Seasonal & Local	Avoid: Imported exotic ingredients (blueberries, kale, macadamia)	Eat: Regional, seasonal produce — cheaper, more bioavailable, aligned with your body's cycle
6. Soak Nuts	Avoid: Unsoaked nuts (enzyme inhibitors impair digestion)	Eat: Soak all nuts 6–8 hours. Discard soaking water. Greatly improves digestibility.
7. Mild Spices Only	Avoid: Garam masala, red chili powder, packaged spice blends	Eat: Fresh green chili, black pepper, curry leaves, coriander, basil, lemongrass, thyme
8. Minimal Cooking	Avoid: High-heat frying, pressure cooking vegetables, prolonged boiling	Eat: Lowest temperature, shortest duration. Raw where possible. Preserve enzymes.
9. No Stimulants	Avoid: Tea, coffee, onions, garlic	Eat: Herbal teas (tulsi, lemongrass, rose), coconut water as the daily beverage

6 Mindful Eating Laws

1. Eat only when you feel GENUINE hunger — not out of boredom, habit, or emotion.
2. Eat only till you are 3/4 FULL — leave space for digestive juices to work.
3. Always eat in a RELAXED, CALM state — never when anxious, angry, or rushed.
4. Give your body 15 MINUTES REST after a heavy meal before resuming work.
5. CHEW THOROUGHLY — each mouthful well-mixed with saliva before swallowing.
6. EXPRESS GRATITUDE before meals — acknowledge the journey food has made to reach you.

Food Combining Rules

Rule 1	In healing phase: only ONE grain or legume per meal. No rice + chapati, no rice + dal.
Rule 2	Mix grains/legumes with 2x more vegetables always.
Rule 3	NEVER mix fruits and cooked food in the same meal.
Rule 4	NEVER mix sweet fruits (mango, banana) with citric fruits (orange, lemon, pineapple).
Rule 5	Do NOT drink water while eating — wait 1 hour before or 2 hours after meals.
Rule 6	Keep meals simple — 2–3 types of food maximum per sitting.

5.3 Three Meal Plans

■ HEALING PLAN — For reversing chronic disease. Maximum 3 months. Then switch.

■ **LIFESTYLE PLAN — Sedentary lifestyle or tendency to gain weight.**

■ **ACTIVE PLAN — Underweight or highly active — athletes, gym-goers.**

Healing Plan — Daily Schedule (Full Detail)

Time	Meal	Options
8:00 AM	Detox Juice	Ash Gourd Juice (400ml) · Coconut Water (fresh, 400ml) · Glowing Green Juice (cucumber, spinach, mint, apple, ginger, lemon) · ABC Juice (Apple, Beetroot, Carrot, Ginger) · Tomato Bathua Juice (winter)
10:00 AM	Fruit Breakfast	Single seasonal fruit (mono-eating) · Mixed seasonal fruits · Blush Smoothie Bowl · Green Smoothie Bowl · Banana Date Shake · Marigold Smoothie · Pure Satvic Salad · Clear Soup (winter) Note: Smoothies max 2-3x/week. Diabetics: guava, papaya, pear, apple only — no smoothies.
1:00 PM	Grain Lunch	Satvic Roti (50% wheat + 50% vegetable) · Millet Roti · Satvic Sabzi · Satvic Khichadi (1 cup rice + 5 cups veg) · Satvic Daliya · Millet Upma · Spinach Cheela · Golden Lentil Bowl · Barley Bowl Rule: ONE grain/legume only + 2-3x more vegetables
4:00 PM	Optional Mid-Meal	Fresh juice · Herbal tea (lemongrass, cinnamon, cardamom) · 6–8 soaked nuts · Fresh coconut slices Skip if not genuinely hungry.
7:00 PM	Veggie Dinner (NO GRAINS in healing)	SALADS: Carrot Raisin · Cheesy Salad (cashew cheese) · Thai Papaya · Zesty Beet · Sweet Potato SOUPS: Pumpkin · Papaya Corn · Spinach · Pea Carrot · Broccoli Potato · Tomato · Carrot Cumin Underweight: add small grain/legume portion

■ Fasting window: maintain 14–16 hours between dinner (7 PM) and morning juice (8–9 AM). This window is essential for cellular repair (autophagy) and hormone regulation.

5.4 Nutrition Reference Guide

Complete Protein Sources (Plant-Based)

Food	Protein	Food	Protein
Cooked lentils	18g / cup	Cooked chickpeas	15g / cup

Cooked kidney beans	15g / cup	Homemade tofu	20g / cup
Green peas	8g / cup	Sprouted moong (raw)	7g / cup
Peanut butter (2 tbsp)	8g	Almonds (handful)	6g
Cashews (handful)	5g	Cooked millets	6g / cup
Brown rice	5g / cup	Whole wheat chapati	3g each

■ WHO recommendation: 0.83g protein per kg body weight for moderately active adults. The healing plan intentionally keeps protein lower to reduce digestive load during recovery.

Calcium Sources (No Dairy Needed)

Poppy seeds	1438mg per 100g
Sesame seeds	1283mg per 100g
Drumstick leaves	314mg per 100g
Finger millet (Ragi)	364mg per 100g
Fenugreek leaves (Methi)	275mg per 100g
Amaranth leaves (Cholai)	245mg per 100g
Quinoa	198mg per 100g

Vitamin D	15–20 min direct sunlight, 3–5x per week. Supplement if blood test shows deficiency.
Vitamin B12	Made by soil organisms — depleted by modern farming. Get blood test. Supplement under physician guidance if deficient. This is one of the few genuine supplementation requirements.

5.5 Sprouts, Nut Milks & Special Occasions

Sprouts Guide

Sprouts are 10–30x more nutritious than the full-grown vegetable — the most concentrated living food available. The sprouting process activates enzymes, increases vitamins, and neutralises anti-nutrients.

Seed	Days to Sprout	Yield
Moong	3 days	0.5 cup seeds → 2 cups sprouts
Green Lentils	3 days	0.5 cup → 2 cups
Chickpeas	3 days	0.5 cup → 1.5 cups
Fenugreek	3 days	1 tbsp → 1.5 cups

Alfalfa	5 days	1 tbsp → 1 cup
Clover	5 days	1 tbsp → 1 cup
Radish	5 days	2 tbsp → 1 cup

Sprout Salad Ratio: 30% sprouts + 30% vegetables + 30% leafy greens + 10% toppings (coconut, nuts, seeds, dressing)

Nut Milks

Coconut Milk	1 cup fresh coconut + 2 cups water. Blend, strain. Use immediately or refrigerate max 1-2 days.
Almond Milk	1/4 cup soaked almonds (6hrs) + 1 cup water. Blend, strain.
Sesame Milk	1/4 cup soaked sesame seeds (6hrs) + 1 cup water. Blend, strain.
Critical Rules	Always make fresh. Never buy packaged. Soak minimum 6hrs. Discard soaking water. NEVER cook coconut milk directly on flame — add after switching off heat.

Special Occasion Recipes

Festival Drinks	Coconut Chaas (mint, lemon, cumin, coconut milk) · Thandai (almonds, dates, saffron, cardamom) · No-Coffee Cold-Coffee (roasted date seeds as substitute)
Desserts	Satvic Kheer (kodo millet + almond milk + jaggery) · Satvic Gajar Halwa (carrots + coconut milk + jaggery) · Peanut Butter Ice Cream (frozen banana + peanut butter) · Lemon Cheesecake (cashew base — no dairy, no refined sugar) · Satvic Ladoo (coconut butter + jaggery)
Main Course	Thai Curry with Brown Rice (homemade Thai paste + seasonal vegetables + coconut milk)

PART VI

ETHICAL FRAMEWORK, LIMITATIONS & ROADMAP

6.1 Ethical Guidelines

Informed Consent	All data collection (video, audio, text) requires explicit opt-in consent before any session begins.
Data Privacy	No persistent storage of biometric data by default. Full deletion option provided. No third-party sharing.
Professional Disclaimer	Every session begins with the disclaimer that Humorix is not a licensed therapist and cannot provide medical diagnoses. Shown once per session.
Bias Mitigation	Facial recognition models are evaluated across skin tone, age, gender, and cultural background. Ongoing audits for demographic performance gaps.
Crisis Escalation	Non-negotiable: when crisis signals are detected, resources are provided immediately and unconditionally — before any other content.
No Diagnosis	Humorix never outputs a clinical diagnosis. It identifies risk patterns and recommends professional evaluation.
Transparency	The system always explains what it detected and why it responded as it did. No black-box outputs.

6.2 Known Limitations

Demographic Accuracy Gaps	FER models trained predominantly on lighter skin tones and Western subjects. Active work needed on South Asian, African, and East Asian training data.
Environmental Dependency	Facial detection degrades significantly in poor lighting, unusual angles, or occluded faces (glasses, masks, camera angle).
Emotional Complexity	Current models classify into discrete categories — real emotions are continuous, mixed, and context-dependent in ways current architectures simplify.

No Longitudinal Memory	Humorix does not currently maintain memory across sessions — each session starts fresh, limiting long-term progress tracking.
Language Limitations	NLP models currently perform best in English. Multilingual and Indian language support (Hindi, Tamil, Telugu, etc.) is limited.
Cultural Calibration	Emotion expression norms vary significantly across cultures — a model trained without cultural calibration will misread emotion in cross-cultural contexts.
Clinical Validation	No large-scale randomised controlled trials yet. Clinical validation studies are planned but not yet published.

6.3 Future Roadmap

PERSONALISED AI	WEARABLES	EMOTIONAL MEMORY	ADAPTIVE UI	CLINICAL TRIALS	MULTILINGUAL

Personalised AI Models	Fine-tune emotion models per individual over time using federated learning — without transmitting raw biometric data to servers.
Wearable Integration	Connect with heart rate monitors, GSR (galvanic skin response) sensors, and sleep trackers to enrich physiological signal data.
Emotional Memory System	Cross-session context: track mood patterns, progress, and trigger awareness over weeks and months.
Real-Time Adaptive UI	Interface dynamically adjusts based on detected emotional state — slower text for anxiety, softer colours for distress, energetic visuals for ADHD engagement.
Clinical Validation Studies	Partner with psychiatric institutions (NIMHANS and equivalents) to run controlled trials measuring efficacy vs. baseline care.
Multilingual Support	Full Hindi, Tamil, Telugu, Bengali, and Marathi NLP models for Indian population reach.
Satvic Integration Depth	Personalised meal plan generation based on detected emotional state + health profile + seasonal availability.

HUMORIX AI — Emotionally intelligent, ethically grounded, holistically designed.

Detect what humans feel. Validate without judgment. Intervene with evidence. Nourish the whole being.

DETECT → VALIDATE → INTERVENE → TRACK → ESCALATE